

Beginner Level : 5 Problems

Problem 1: Smart Light

A smart light turns **ON** if the room is bright enough.

You are given two integers:

- `brightness`
- `threshold`

If `brightness` $\geq$ `threshold`, print **ON**. Otherwise, print **OFF**.

Input Format

Two integers: brightness and threshold

Output Format

ON

or

OFF

Sample Input

75 60

Sample Output

ON

Problem 2: Even or Odd

You are given an integer `N`.

Determine whether the number is **Even** or **Odd**.

Input Format

One integer N

Output Format

Even

or

Odd

Sample Input

Sample Output

Odd

Problem 3: Print Numbers

You are given an integer N .

Print all numbers from 1 to N , one number per line.

Input Format

One integer N

Sample Input

5

Sample Output

1
2
3
4
5

Problem 4: Count Even Numbers

You are given N integers.

Count how many of them are **even**.

Input Format

- First line contains N .
- Second line contains N integers.

Sample Input

6
2 7 8 11 14 5

Sample Output

3

Explanation:

The even numbers are 2, 8, 14.

Problem 5: Sum of Odd Numbers

You are given an integer N .

Find the sum of all **odd numbers** from 1 to N .

Input

10

Output

25

Explanation:

The odd numbers are:

$$1 + 3 + 5 + 7 + 9 = 25$$

Hint:

You can use `continue` to skip the even numbers.

Medium Level : 3 Problems

Problem 6: Smallest and Largest

You are given N integers.

Find the **smallest** and **largest** number among them.

Do not use Python's built-in `min()` or `max()` functions.

Input Format

- First line contains N .
- Second line contains N integers.

Sample Input

5
12 7 25 3 18

Sample Output

3 25

Problem 7: Positive, Negative, Even and Odd

You are given N integers.

Count:

1. How many numbers are positive
2. How many numbers are negative
3. How many numbers are even
4. How many numbers are odd

Zero is **neither positive nor negative**, but zero is **even**.

Input Format

First line: N

Second line: N integers

Sample Input

```
7
-5 0 8 -2 7 11 4
```

Sample Output

```
Positive: 4
Negative: 2
Even: 4
Odd: 3
```

Problem 8: Reverse Digits

You are given a positive integer **N**.

Print its digits in **reverse order**.

Input

```
12345
```

Output

```
5 4 3 2 1
```

For example:

Input: 908

Output: 8 0 9

Hint:

Use:

`number % 10`

to get the last digit, and

`number // 10`

to remove the last digit.

Hard Level : 2 Problems

Problem 9: Find the First Number Greater Than 100

You are given N integers.

Find the **first number greater than 100**.

As soon as you find the first number greater than 100, stop checking the remaining numbers.

If there is no number greater than 100, print:

Not Found

Input Format

First line: N

Second line: N integers

Sample Input

```
7
20 45 80 105 150 200 300
```

Sample Output

```
105
```

Important:

Even though 150, 200, 300 are also greater than 100, only 105 should be printed because it is the **first one**.

Problem 10: Number Analysis

You are given N positive integers.

For each number:

- If the number is **even**, ignore it and move to the next number.
- If the number is **odd**, calculate the **sum of its digits**.
- If the digit sum is greater than 20, stop processing any more numbers.
- Otherwise, print the number and its digit sum.

At the end, print how many odd numbers were successfully processed.

Sample Input

```
6
```

12 135 47 91 999 25

So the output will be:

135 9

47 11

91 10

3